Euclid's Elements — Book I
Proposition 24

Formal Reconstruction — Technical Documentation

Jurek Różański
Military University of Technology, Warsaw, Poland
jerzy.rozanski@wat.edu.pl

CGJteamLab

A greater included angle gives a greater third side

1 Introduction

Euclid's Proposition I.24 is the classical hinge theorem. Two triangles have two corresponding pairs of congruent sides. If the included angle of the first triangle is greater than the included angle of the second, then the side opposite the greater angle is greater than the corresponding side of the second triangle.

In the notation of the current formal development, the hypotheses are

$$AB \cong DE, \qquad AC \cong DF,$$

and

$$\angle EDF < \angle BAC.$$

The conclusion is

$$EF < BC.$$

The reconstruction is entirely synthetic. No numerical length function, coordinate system, or numerical angle measure is introduced. Proposition I.24 is nevertheless substantially more delicate than its statement suggests. The traditional diagram suppresses a genuine positional issue, and the Lean reconstruction makes the required order information explicit.

There are two distinct three-way analyses in the documentation, and they must not be identified:

- the *historical* three configurations concern the position of the classical auxiliary points relative to the line GF;
- the *formal* three cases concern only the order of the points H and Y on one ray from A.

2 The Classical Configuration

Let $\triangle ABC$ and $\triangle DEF$ be nondegenerate triangles with

$$AB \cong DE, \qquad AC \cong DF, \qquad \angle EDF < \angle BAC.$$

Euclid constructs at D an angle congruent to the larger angle at A and lays off a segment DG congruent to AC and hence to DF. Thus

$$\angle EDG \cong \angle BAC, \qquad DG \cong AC \cong DF.$$

By SAS,

$$\triangle ABC \cong \triangle DEG,$$

so that

$$BC \cong EG.$$

It remains to prove $EF < EG$.

The historical difficulty is best seen by displaying all three configurations distinguished in the classical commentary tradition. Following the preserved Book1R discussion, the distinction is made by the positions of D and E relative to the line through G and F.

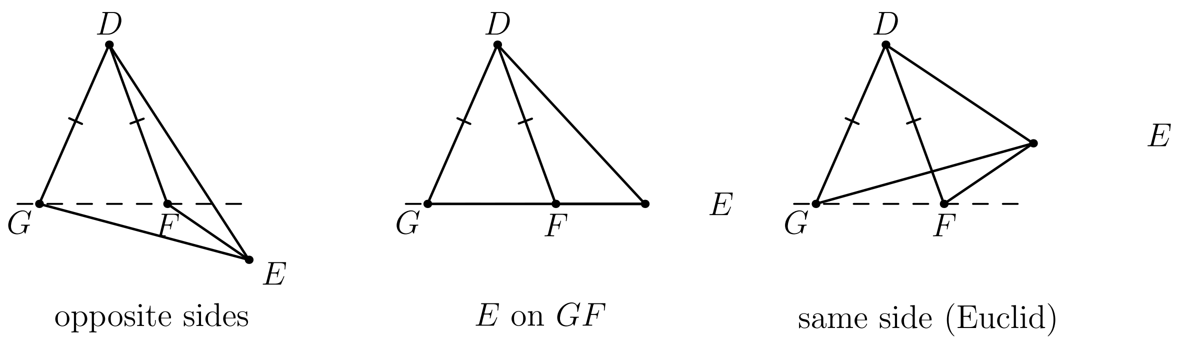

Figure 1: The three configurations distinguished in the classical discussion of Proposition I.24: D and E on opposite sides of the line GF, E on GF, and D, E on the same side. Euclid's printed proof treats the third. The equal marks on DG and DF record the isosceles triangle used in the classical argument. These are historical configurations, not the three Lean hinge branches.

3 Classical Proof and the Configuration Gap

Euclid's argument proceeds as follows. Since

$$AB \cong DE, \qquad AC \cong DG, \qquad \angle BAC \cong \angle EDG,$$

Proposition I.4 gives

$$BC \cong EG.$$

Also $DF \cong DG$, so triangle DFG is isosceles. Proposition I.5 therefore supplies the corresponding equality of its base angles. In the configuration displayed by Euclid, this angle equality together with the position of E yields a strict comparison between the two angles of triangle EFG. Proposition I.19 then gives

$$EF < EG.$$

Since $EG \cong BC$, one obtains $EF < BC$.

The classical dependency pattern is therefore

```
angle EDF < angle BAC
        |
construct G: DG ~= AC, angle EDG ~= angle BAC     I.23
        |
       I.4
        |
     EG ~= BC
        |
     DG ~= DF
        |
       I.5
        |
angle comparison in triangle EFG
        |
       I.19
        |
     EF < EG
        |
     EF < BC
```

The problem is not the congruence part of this argument. The problem is the unproved assumption about the relative position of the auxiliary points. Ancient and later commentary already recognized the omitted configurations; the Hilbert reconstruction supplies an axiomatic language in which such position information cannot be inferred silently from one preferred picture.

In Euclid's third configuration the useful angle-comparison chain may be written schematically as

$$\angle EFG > \angle DFG = \angle DGF > \angle EGF.$$

The preserved Book1R PDF displayed this same configuration a second time with explanatory coloured angle regions. No new point, line, betweenness relation, or construction appears there, so the audited version keeps the comparison in the text rather than retaining a duplicate figure.

4 Strict Angle and Segment Comparison

Two synthetic order relations are central to the reconstruction. `HilbertAngleLess` expresses strict comparison of angles without a numerical angle measure. Informally, $\alpha < \beta$ means that a congruent copy of α occurs as a proper interior subangle of β.

Similarly, `HilbertSegmentLess` expresses strict comparison of segments through betweenness and congruence, without introducing a real-valued length function. Proposition I.24 bridges these two order theories: its hypothesis is a strict angle inequality and its conclusion is a strict segment inequality.

5 Formal Architecture of the Current Lean Proof

The current production file imports

```lean
import CGJteamLab.HilbertBookZero
import CGJteamLab.Proposition19
import CGJteamLab.Proposition23
```

and contains the proposition-specific theorem chain

```
euclid_proposition_24_construct_aux
euclid_proposition_24_sas_aux
euclid_proposition_24_hinge_case1
euclid_proposition_24_hinge_case3
euclid_proposition_24_hinge_aux
euclid_proposition_24
```

All these declarations use the Hilbert incidence and congruence typeclasses. No Euclidean-plane typeclass is required by I.24.

5.1 The Interior-Ray Construction

The input

$$\angle EDF < \angle BAC$$

is already a witness-bearing synthetic statement. The theorem `euclid_proposition_24_construct_aux` extracts a point X whose ray from A meets the open segment BC. Unpacking that meeting relation provides a point Y with

$$B - Y - C, \qquad \mathrm{SameRay}(A; X, Y).$$

The proof then lays off on the ray AX a point H satisfying

$$\mathrm{SameRay}(A; X, H), \qquad AH \cong AC, \qquad \angle BAH \cong \angle EDF.$$

The point X is an important representation witness, but it is not given a separate diagram. The formal statements do not determine a special order of X relative to H and Y, and placing it arbitrarily on a figure would add positional information not supplied by the theorem.

5.2 The SAS Construction

The theorem `euclid_proposition_24_sas_aux` packages the congruence part of the argument. From

$$AH \cong AC \cong DF, \qquad AB \cong DE, \qquad \angle BAH \cong \angle EDF,$$

SAS applied to $\triangle ABH$ and $\triangle DEF$ gives

$$BH \cong EF.$$

The original target $EF < BC$ is therefore reduced to the hinge comparison

$$BH < BC.$$

5.3 The Hinge Configuration

The interior ray through H meets the open segment BC at Y. Same-ray information does not determine the order of H and Y on the ray from A. The current proof invokes `euclid_proposition_22_sameRay_trichotomy` and obtains exactly

$$A - H - Y, \qquad H = Y, \qquad A - Y - H.$$

This is the central geometric case split of the current Lean proof. It is not the historical three-way split of Figure 1.

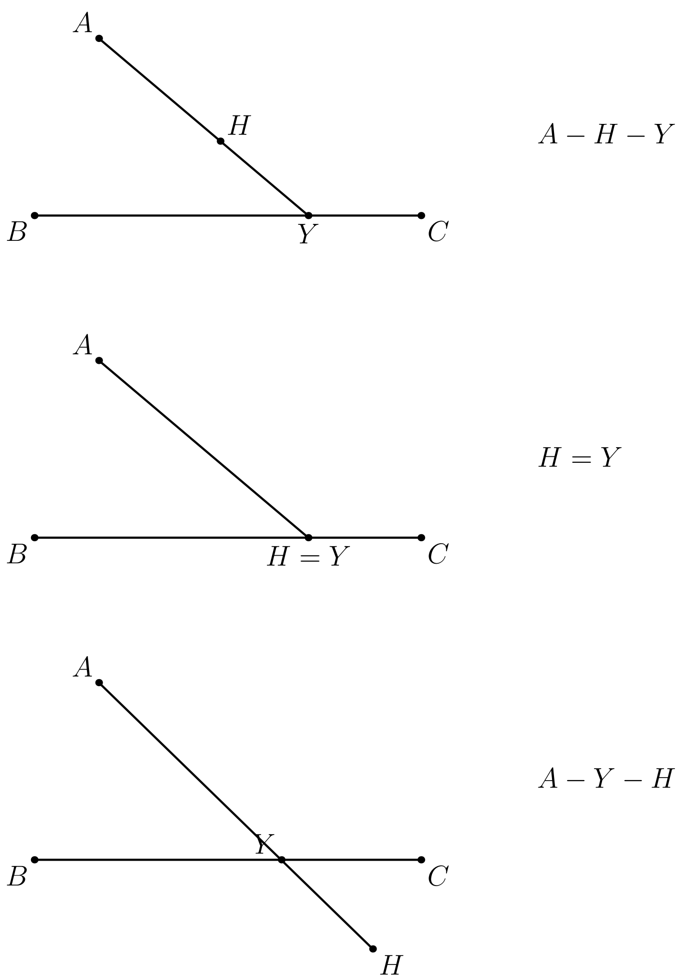

Figure 2: The three order configurations used by `euclid_proposition_24_hinge_aux`. In all three panels Y lies in the open segment BC; the split concerns only the order of H and Y on their common ray from A. The earlier SAS helper also supplies $AH \cong AC$ and $BH \cong EF$, but those congruence facts do not create additional order configurations.

5.4 Case 1: $A - H - Y$

This is the longer nontrivial branch. After reconstructing the required noncollinearity facts, the proof uses Proposition I.16 to obtain strict exterior-angle comparisons. Since

$$AH \cong AC,$$

triangle AHC is isosceles, allowing the relevant angles at H and C to be transported across congruence. The incidence of the ray through Y with the opposite side is converted into strict angle order by `hilbert_interior_angle_less`. After the required same-ray normalizations and transitivity steps, the branch obtains

$$\angle BCH < \angle BHC.$$

The current code then calls `euclid_proposition_19` directly and concludes

$$BH < BC.$$

5.5 Case 2: $H = Y$

The middle branch is genuinely different and much simpler. After substituting $H = Y$, the already available betweenness

$$B - Y - C$$

becomes $B - H - C$. The witness-based definition of strict segment order immediately supplies

$$BH < BC.$$

No angle comparison is required.

5.6 Case 3: $A - Y - H$

This branch has different incidence geometry. At C, the ray through Y meets the open segment HA, producing a proper interior-subangle comparison. The betweenness $B - Y - C$ identifies the relevant ray at C; $AH \cong AC$ transfers the comparison across the isosceles triangle AHC. At H, the ray through Y meets the open segment CB, giving a second interior-angle comparison. Since $H - Y - A$, the ray HY agrees with the ray HA. The transported inequalities again yield

$$\angle BCH < \angle BHC,$$

and a direct application of Proposition I.19 gives $BH < BC$.

5.7 The Hinge Helper and Completion

The theorem `euclid_proposition_24_hinge_aux` packages the complete order analysis:

```
ray through H meets BC at Y
        |
same-ray trichotomy for A,Y,H
        |
 +------+-------+
 |      |       |
A-H-Y  H=Y    A-Y-H
 |      |       |
case1  direct  case3
 +------+-------+
        |
     BH < BC
```

The public theorem then has the short endpoint architecture

```
euclid_proposition_24_sas_aux
        |  BH ~= EF
        v
euclid_proposition_24_hinge_aux
        |  BH < BC
        v
bookZero_32_lessThanCongruence2
        |
        v
     EF < BC
```

Thus the final strict segment comparison is transported explicitly by the Book Zero theorem

```
bookZero_32_lessThanCongruence2
```

6 Diagrammatic Audit

The preserved Book1R PDF contained three figures for I.24:

1. a three-panel map of the historical classical configurations;
2. the Euclid same-side configuration repeated with explanatory angle regions;
3. a three-panel map of the formal hinge trichotomy.

The audited version retains all six genuine geometric states but reduces the chapter to two figures. Figure 1 records the three historically distinct positions. Figure 2 records the three formally distinct orders of H and Y.

The former Figure 24.2 is removed because it repeated the third panel of the historical map and changed only the logical interpretation of angles. No new construction or order relation appeared there. The comparison is therefore kept in prose rather than as a separate float.

No additional figure is introduced for the witness X, for SAS, for angle-congruence transport, or for the final call to `bookZero_32_lessThanCongruence2`. Those steps add no new proposition-specific geometric state. This implements the audit rule: *a figure should record geometry, not tactic state.*

7 Relationship with Earlier Propositions

7.1 Proposition I.5

The isosceles base-angle theorem is used after $AH \cong AC$. It is the main congruence bridge between the angles at H and C in the formal hinge triangle. In the classical proof I.5 is likewise applied to the isosceles triangle DGF.

7.2 Proposition I.16

Case 1 uses the exterior-angle theorem to obtain strict angle comparisons which are not available directly from the interior-ray configuration.

7.3 Proposition I.19

Both nontrivial formal branches end with

$$\angle BCH < \angle BHC,$$

and call `euclid_proposition_19` directly to conclude $BH < BC$. In the classical proof I.19 converts the angle comparison in $\triangle EFG$ into $EF < EG$.

7.4 Proposition I.23

The production module imports `Proposition23`, and I.23 belongs to the classical dependency picture as the construction of an angle equal to the given larger angle. The current I.24 proof body, however, does not call `euclid_proposition_23` directly; its auxiliary angle is obtained through the more general witness machinery already packaged in `HilbertAngleLess`.

8 Hilbert and Book Zero Components Used

The main reusable infrastructure is:

- `HilbertAngleLess`, carrying the interior-ray witness of the strict input inequality;
- `HilbertSegmentLess`, the witness-based target relation;
- `HilbertSameRay` and the I.22 same-ray trichotomy;
- `hilbert_interior_angle_less`;
- angle-congruence and angle-order transport;
- Book Zero collinearity, order, and strict-segment transport lemmas.

The current module itself contains no declaration of `axiom` or `sorry`. This documentation audit did not rerun `lake build` or `#print axioms`; therefore no new build or transitive axiom-audit claim is made here.

9 Historical Note

Proposition I.24 is a useful example of the distinction between a classical geometric argument and a complete axiomatic treatment of its diagram. The traditional text presents one configuration; ancient commentary already supplied arguments for the omitted configurations, and later editors reorganized or restricted the construction in different ways. Hilbert's role is different: incidence, betweenness, ray identity, and separation must be stated and justified instead of being silently read from a preferred picture.

Accordingly, the three-way split in the present Lean proof is not a proof-assistant artifact. It is a precise order-theoretic analysis of the auxiliary hinge configuration. At the same time, it must not be confused with the three historical configurations in the classical discussion.

10 Dependency Summary

The formal dependency structure can be summarized as

```
Hilbert incidence + order + congruence
        |
HilbertAngleLess / HilbertSegmentLess / HilbertSameRay
        |
angle construction + interior-ray theory + I.22 trichotomy
```

```
        |
I.5 + I.16 + I.19 + Book Zero transport
        |
construct_aux -> sas_aux
        |
hinge_case1 / direct middle case / hinge_case3
        |
hinge_aux
        |
bookZero_32_lessThanCongruence2
        |
euclid_proposition_24
```

No numerical metric, coordinate system, numerical angle measure, or new hinge axiom is introduced by the proposition.

11 Conclusion

Proposition I.24 is one of the clearest examples of what formal reconstruction can reveal about a classical synthetic proof. The metric core is short: the SAS construction gives $BH \cong EF$, and the hinge comparison gives $BH < BC$. The real difficulty is positional: the hypotheses do not force a single order of the auxiliary point H and the intersection point Y.

The audited documentation therefore keeps exactly the geometry that matters. It retains a three-panel historical map and a separate three-panel formal hinge map, while removing the duplicate coloured angle figure. The resulting two-figure presentation preserves all genuine geometric states and separates the classical and formal DAGs explicitly.